

# Soil Pollution

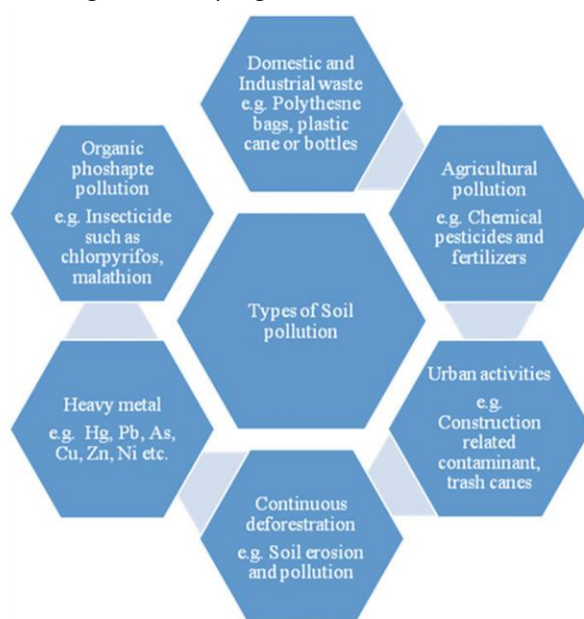
## Soil Pollution

Soil contamination, soil pollution, or land pollution as a part of land degradation is caused by the presence of xenobiotic (human-made) chemicals or other alteration in the natural soil environment. It is typically caused by industrial activity, agricultural chemicals or improper disposal of waste. Contamination is correlated with the degree of industrialization and intensity of chemical substance. The concern over soil contamination stems primarily from health risks, from direct contact with the contaminated soil, vapour from the contaminants, or from secondary contamination of water supplies within and underlying the soil.



## Types of Soil Pollutants

All soils, whether polluted or unpolluted, contain a variety of compounds (contaminants) which are naturally present. Such contaminants include metals, inorganic ions and salts (e.g. phosphates, carbonates, sulfates, nitrates), and many organic compounds (such as lipids, proteins, DNA, fatty acids, hydrocarbons, PAHs, alcohols, etc.). These compounds are mainly formed through soil microbial activity and the decomposition of organisms (e.g., plants and animals). Additionally, various compounds get into the soil from the atmosphere, for instance with precipitation water, as well as by wind activity or other types of soil disturbances, and from surface water bodies and shallow groundwater flowing through the soil. When the amounts of soil contaminants exceed natural levels (what is naturally present in various soils), pollution is generated. This can occur through anthropogenic (man-made) causes as well as natural causes.



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## Anthropogenic Soil Pollutants

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Anthropogenic (man-made) soil pollution originates in several types of processes, some deliberate (industrial) and others, accidental. Human-caused soil pollution can work in conjunction with natural processes to increase the toxic contamination levels in the soil.

- Accidental spills and leaks during storage, transport or use of chemicals (e.g., leaks and spills of gasoline and diesel at gas stations)
- Foundry activities and manufacturing processes that involve furnaces or other processes resulting in the possible dispersion of contaminants in the environment;
- Mining activities involving the crushing and processing of raw materials, for instance, heavy metals, emitting toxic substances;
- Construction activities
- Agricultural activities involving the diffusion of herbicides, pesticides and/or insecticides and fertilizers;
- Transportation activities, releasing toxic vehicle emissions
- Chemical waste dumping, whether accidental or deliberate - such as illegal dumping;
- The storage of waste in landfills, as the waste products may leak into groundwater or generate polluted vapours
- Cracked paint chips falling from building walls, especially lead-based paint.

Construction sites are the most important triggers of soil pollution in urban areas, due to their almost ubiquitous nature. Almost any chemical substance handled at construction sites may pollute the soil. However, the higher risk comes from those chemicals that can travel more easily through the air as fine particulate matter. The chemicals that travel as particulate matter are more resistant to degradation and bioaccumulate in living organisms, such as PAHs.

Additionally, construction dust may easily spread around through the air and is especially dangerous because of its lower particle size (less than 10 microns). Such construction dust can trigger respiratory illnesses such as asthma and bronchitis, and even cancer. Moreover, the sites that involve the demolition of older buildings can release asbestos, a toxic mineral that can act as a poison in soil. Asbestos particles can be redistributed by the wind.



## Natural Pollutants

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Apart from the rare cases when a natural accumulation of chemicals leads to soil pollution, natural processes may also have an influence on the human released toxic chemicals into the soil, decreasing or increasing the pollutant toxicity and the level of contamination of the soil. This is possible due to the complex soil environment, involving the presence of other chemicals and natural conditions which may interact with the released pollutants.

The following are some of the natural processes leading to soil pollution:

- There may be a natural accumulation of compounds in soil due to imbalances between atmospheric deposition and leaking away with precipitation water (e.g., concentration and accumulation of perchlorate in soils in arid environments).
- Natural production in soil may also occur under certain environmental conditions (e.g., natural formation of perchlorate in soil in the presence of a chlorine source, metallic object and using the energy generated by a thunderstorm).
- Sewer lines may leak from into the subsurface (e.g., adding chlorine could generate trihalomethanes such as chloroform).

## Effects of Soil Pollution

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Soil pollution is devastating to the environment and has consequences for all forms of life that encounter it. The toxic substances that are deposited on the earth's surface harm our health and well-being and affect food, water and air quality directly and indirectly. The most important effects of soil pollution are indicated below:

### 1. Effect on Health of Humans

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Considering how soil is the reason we are able to sustain ourselves, its contamination has major consequences on our health. Crops and plants grown on polluted soil absorb much of the pollution, which is then passed on to us. This could result in the sudden surge of certain illnesses. Long term exposure to such soil can affect the genetic make-up of the body, causing congenital illnesses and chronic health problems that cannot be cured easily. In fact, it can sicken the livestock to a considerable extent and cause food poisoning over a long period of time. The soil pollution can even lead to widespread famines if the plants are unable to grow in it.

There are three main routes where soil can directly enter the body: inhalation, eating, or through skin contact.

#### Inhalation

This route mostly affects people, like workers, who are continually working with soil or those who reside nearby such areas. These types of environments have fine dust particles floating around, which can be inhaled and eventually absorbed by the body.

#### Eating

Adults consume soil through accidental ingestion. An example of this is when the ingested food, like vegetables, still has some soil attached to it. However, in some parts of the world, the soil is deliberately consumed due to cultural reasons. On the other hand, children, especially those under the age of three, are at high risk for soil contamination exposure as they tend to eat soil while playing outdoors. In addition, their biological makeup is more likely to absorb more of the toxic chemicals than that of an adult.

#### Skin Contact

Also known as "dermal absorption" or "cutaneous absorption," this route is most applicable to volatile organic compounds. However, some heavy metals do cause skin contact problems.

### 2. Effect on Growth of Plants

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The ecological balance of any system gets affected due to the widespread contamination of the soil. Most plants are unable to adapt when the chemistry of the soil changes so radically within a short period of time. Fungi and bacteria found in the soil that bind it together, begin to decline,

which creates an additional problem of soil erosion. The fertility of the soil thus slowly diminishes, making the land unsuitable for agriculture and for any local vegetation to survive. Soil pollution causes large tracts of land to become hazardous to health.

### **3. Decreased Soil Fertility**

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The toxic chemicals present in the soil can decrease soil fertility, which in turn affects its yield. The contaminated soil is then used to produce fruits and vegetables, which lacks quality nutrients and may contain some poisonous substance that can cause serious health problems to the people consuming them.

### **4. Toxic Dust**

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The emission of toxic and foul gases that emanates from landfills pollutes the environment, causing serious health problems to people. Another side effect is the unpleasant smell causes that causes inconvenience to people living in the vicinity.

### **5. Changes in Soil Structure**

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Soil pollution can lead to the death of several soil organisms (e.g., earthworms) in the soil that can alter the soil structure. In addition, it could also force animals to migrate to other places in search of food.

### **6. Poisoning of the Underground Water Table**

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Soil pollution can also result in contamination of the underground water table. This water, being present beneath soil layers, the toxins can easily percolate slowly and steadily into the water table. Since this is the water that is available for consumption and usage through wells and tube wells, it causes a lot of ill effects on our health. Diseases like arsenic poisoning, food poisoning and others are caused due to the prolonged consumption of this toxic underground water. These diseases could also prove to be quite fatal.

## **Possible Solutions to Soil Pollution**

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With a global population that is projected to exceed 9 billion by 2050, our current and future food security hinges on our ability to increase yields and food quality using the soils that we already have available today. Soil pollution negatively impacts us all, and has been identified as one of the main threats to soil functions worldwide. We need to be aware of the causes of soil pollution so that we can create and implement solutions. Soil protection and conservation starts with us. Making sustainable food choices, properly recycling dangerous materials like batteries, composting at home to reduce the amount of waste that enters landfills or managing antibiotic waste more responsibly, are just a few examples of how we can be part of the solution. On a larger scale, we need to promote sustainable agricultural practices in our communities. Soil pollution is a complex problem that ought to be solved. It is essential that we all realize how important soil is for us. The earlier we realize, the better we will be able to solve this problem. It is a complex problem, and thus, it requires everyone, from an individual to the government, to work in complete unison. Listed below are a few things that could help in reducing soil pollution.

### **1. Reduced Use of Chemical Fertilizers**

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Chemical fertilizers do more harm than good. While proper amounts could enhance the fertility of the soil, excess of it actually poisons the soil. The excess of chemical fertilizers could pollute the soil in several ways. It could mess with the pH levels of the soil. It could also destroy the good microorganisms in the soil. Not only that, but the runoffs from such soils also cause water pollution as well. Thus, using chemical fertilizers is like a double-edged sword.

## **2. Promotion of Reforestation and Afforestation**

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One of the major causes of soil pollution is soil erosion due to deforestation. It is natural that with the ever-growing population, the humankind needs more and more space to expand their civilization. Often it is achieved at the cost of the health of the soil. To prevent this from happening, reforestation of a deforested area should be promoted. Also, afforestation should be promoted in the barren lands. The roots of the plants bind the soil particles together and even capture good microorganisms in the soil. It also ensures the maintenance of the underground water table.

## **3. Recycle and Reuse Products**

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These steps not only reduce waste generation but also ensure that soil pollution is reduced. At present, plastic forms a significant portion of the generated wastes. More often than not, these wastes are buried in landfills. In these landfills, these plastics and other materials decompose slowly and release toxic materials into the soil. These toxic substances are very harmful to the health of the soil and are a major source of soil pollution. By reusing and recycling things, we can ensure that lesser wastes are dumped in these landfills, and this, in turn, would reduce soil pollution.

## **4. Get the Locals Involved**

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In order to ensure that a problem like soil pollution is solved, it is essential that every individual must get involved. It is with their involvement that things can work out better. Awareness programs could be designed so that people understand soil pollution better. If people are aware, they will help even subconsciously.

## **5. Promote Use of Natural Manure**

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Natural manure is one of the best sources of nutrients for the soil. It is harmless and completely organic. It adds essential nutrients to the soil and restores the health of the soil. It has no harmful by-products that could harm the soil or the environment in any way.